Amazon Product Analysis

Pricing, Discounts, Ratings, Customer Feedback

Exploratory Data Analysis Report

This report summarizes the cleaning and exploratory analysis carried out on an Amazon product dataset using Microsoft Excel. It documents the main patterns observed, the dashboard content, and important data-quality limitations to resolve before presenting the results as final business conclusions.

1. Executive Summary

The analysis examined product records, categories, listed discounts, prices, ratings, and rating counts. The dataset contains 1,465 records and 1,351 unique product IDs, indicating repeated product IDs that need to be understood before records are removed or totals are interpreted. USB cables are the most represented category in the category-count results shared, with 233 records. Average ratings are generally positive in the reviewed summaries, but some PivotTables returned conflicting overall figures; these should be reconciled before publishing a single headline rating.

2. Dataset Overview

Measure	Reported result	Interpretation
Total records	1,465	Rows in the dataset, excluding the header.
Unique product IDs	1,351	Distinct product identifiers.
Additional records beyond one per unique ID	114	Repeated-ID records; not necessarily exact duplicate rows.
Distinct category values	211	Detailed category paths are represented as separate values.
Unique reviewer IDs	1,194	Distinct reviewer identifiers reported during profiling.
Blank cells	None reported	A blank-cell check found no blanks; whitespace-only values may still need checking.

3. Data Cleaning and Validation

Pricing and discount validation

The dataset includes discounted price, actual price, and listed discount percentage. Several records had discounted prices greater than actual prices, and the listed discount percentages did not consistently agree with the discount calculated from the two price fields. A helper column was used to test possible decimal-place corrections and to calculate discounts from the cleaned discounted price.

Important limitation: a formula-based decimal correction is an assumption, not proof of the correct price. Keep the original columns, document the correction rule, and manually verify suspicious records against a reliable source wherever possible. The validation comparison previously returned 43 matches and 1,422 checks using the original listed discount percentage, indicating substantial inconsistency.

Repeated product IDs

Repeated product IDs were observed. They should not automatically be deleted: repeated IDs can reflect repeated product records, variants, or other dataset structure. Exact duplicate rows should be distinguished from repeated IDs before deduplication. Rating counts should also not be summed across repeated records without checking whether the same count is repeated.

4. Exploratory Data Analysis Findings

4.1 Category representation

Rank	Category (short name)	Record count	Average rating
1	USB Cables	233	4.15
2	Smartwatches	76	4.03
3	Smartphones	68	4.10
4	Smart Televisions	63	4.21
5	In-Ear Headphones	52	3.90
6	Remote Controls	49	3.80
7	Mixer Grinders	27	4.01
8	Dry Irons	24	4.13
9	Computer Mice	24	4.29
10	HDMI Cables	24	4.25

USB cables are the most represented category among the category results shared, with 233 records. Computer mice (4.29) and HDMI cables (4.25) have the highest average ratings among these ten categories, while remote controls have the lowest average rating (3.80). These counts are record counts, not sales or confirmed unique product counts. The category PivotTable's Grand Total count was reported as 640 rather than 1,465, so the source range and filters must be checked before treating this table as a complete dataset-wide ranking.

4.2 Ratings

A rating PivotTable grouped records into ranges: 1,110 records in the 4–5 range (average 4.22), 348 in the 3–4 range (average 3.73), 6 in the 2–3 range (average 2.57), and 1 below 2 (rating 1). That PivotTable reported an overall average rating of 4.09 across 1,465 records. A separate category-rating PivotTable showed a Grand Total average of 4.48, which conflicts with the 4.09 result. Verify the source range, filters, aggregation, and rating field before publishing a single overall average.

4.3 Listed discounts

The original listed discount percentage had an average of approximately 48%, a median of 50%, a maximum of 94%, and a minimum of 0%. Because listed discounts did not consistently match discounts recalculated from prices, the report should clearly distinguish the original listed discount from any recalculated discount.

4.4 Discount band vs. average rating

Discount band	Average rating	Records
0–9%	4.10	48
10–19%	4.11	62
20–29%	4.11	85
30–39%	4.06	83
40–49%	4.10	113
50–59%	4.12	157
60–69%	4.13	170
70–79%	4.09	254
80–89%	4.08	454
90–100%	4.03	39

Average ratings across discount bands varied only from 4.03 to 4.13. This suggests no strong visible relationship between discount band and average rating in this summary. It does not establish that discounts have no effect; this is an exploratory comparison and does not demonstrate causation.

4.5 Discount band vs. rating count

Discount band	Average rating count	Records
0–9%	9,890.90	48

10–19%	9,734.61	62
20–29%	12,425.98	85
30–39%	12,786.74	83
40–49%	11,712.19	113
50–59%	14,018.76	157
60–69%	11,045.60	170
70–79%	11,646.71	254
80–89%	12,709.38	454
90–100%	5,115.26	39

The 50–59% discount band had the highest average recorded rating count (about 14,019), while the 90–100% band had the lowest (about 5,115). The pattern is uneven and does not show that larger discounts consistently bring more feedback. Rating count is not sales volume, and repeated records may influence the averages.

4.6 Highly rated products with substantial rating counts

Product (short name)	Average rating	Maximum recorded rating count
Swiffer Instant Electric Water Heater Faucet Tap	4.80	53,803
Instant Pot Vortex 2QT Air Fryer	4.80	3,964
Spigen EZ Fit Screen Protector for iPhone 14 Pro Max	4.70	7,779
Sony Bravia 65-inch Smart TV	4.70	5,935
Campfire Spring Chef Prolix Water Heater	4.70	2,591
Multifunctional Electric Egg Boiler/Frying Pan	4.70	2,300
FIGMENT Handheld Milk Frother	4.70	1,729

The Swiffer water heater and Instant Pot air fryer had the highest average rating in this filtered set at 4.80. The Swiffer product also had the largest maximum recorded rating count (53,803). These figures represent recorded ratings, not sales. Using the maximum rating count rather than summing counts helps reduce inflation when the same product's count is repeated across rows, but the underlying records should still be checked.

4.7 Pricing observations

The category price comparison shared in the analysis placed room heaters at the highest average actual price among the displayed categories (2,294.00), followed by rice and pasta cookers (1,923.00) and sandwich makers (1,868.45). These are only the leading categories in the displayed subset; they should not be described as the highest across the full dataset unless the complete PivotTable has been verified.

5. Key Insights

- **Dataset concentration:** USB cables account for the largest category count in the results shared (233 records), substantially more than the next category, smartwatches (76).
- **Ratings are generally positive:** The rating distribution is concentrated in the 4–5 range, but the overall average must be reconciled because PivotTables returned 4.09 and 4.48.
- **Discounts do not align consistently:** Listed discount percentages and price-derived discounts showed substantial mismatches; data validation is essential before pricing conclusions are made.
- **Discount size and rating are similar across bands:** Average ratings by discount band ranged from 4.03 to 4.13, with no strong visible pattern in this summary.
- **Feedback counts peak at mid-range discounts:** The 50–59% discount band had the highest average rating count, while the 90–100% band had the lowest; this is not evidence of a causal effect.
- **Several products combine high ratings with many ratings:** The Swiffer water heater (4.80; max rating count 53,803) and Instant Pot air fryer (4.80; 3,964) led the filtered product list.
- **Counts need careful interpretation:** Record counts and rating counts are not sales, and repeated product IDs may affect category counts and rating-count summaries.

6. Dashboard Components

The Excel dashboard was assembled with KPI cards and EDA charts. Recommended components are:

- KPI cards: total records, unique product IDs, overall average rating, average listed discount, and distinct category values.
- Product rating distribution chart.
- Top 10 categories by record count.
- Average rating by top 10 categories.
- Average rating count by discount band.
- Key Insights text section.

Before sharing the dashboard, ensure all PivotTables use the intended full data range, clear unintended filters, and refresh after any data changes. Specifically resolve the 640 vs. 1,465 category-count discrepancy and the 4.09 vs. 4.48 overall-rating discrepancy.

7. Recommendations

- Reconcile the overall rating calculation and document the final verified value.
- Check PivotTable source ranges, filters, and refresh status; ensure category counts cover the intended dataset.
- Retain original price fields and document every adjustment to discounted prices. Verify unusual values against source data where possible.
- Separate exact duplicate rows from repeated product IDs before deciding whether any rows should be removed.
- Avoid summing rating_count across repeated product records unless the count is confirmed to be additive.
- Use careful wording: record counts are not sales, ratings are not verified purchase counts, and exploratory patterns do not prove causation.
- When comparing categories, show product/record counts alongside average ratings so small groups are not overinterpreted.

8. Conclusion

This project demonstrates practical use of Microsoft Excel for data profiling, price and discount validation, PivotTable analysis, exploratory comparisons, charting, and dashboard development. The analysis suggests that USB cables are the most represented category in the results reviewed, ratings are concentrated in the higher range, and rating counts vary across discount bands without a consistent upward pattern. The strongest next step is to resolve the identified source-range and metric discrepancies and document pricing corrections so the final dashboard is transparent, reliable, and suitable for a portfolio.